

THE UNIVERSITY OF ILLINOIS AT CHICAGO
Department of Chemical Engineering

Midterm #2

Course: CHE 341. Chemical Process Control
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Your score.....

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CHE 341: Chemical Process Control

Instructor: Michael Caracotsios

Submitted by:

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Glossary

Term	Symbol
Controller Gain	K_c
Controller Transfer	G_c
Disturbance Transfer Function	G_D
Error	E
Integral Time Constants	τ_I
Process Gain	K_P
Process Lag	τ_P
Process Response	Y
Process Time Delay	Θ_P
Process Transfer Function	G_P
Process Variable	PV
Proportional (Controller)	P
Proportional – Integral (Controller)	PI
Proportional – Integral – Derivative (Controller)	PID
Set Point	SP
Set Point for Distillate Composition	SP_{XD}
Set Point for Bottoms Composition	SP_{XB}
Set Point for Feed Composition	SP_{XF}

Problem 1: Derivations

The simple feedback control block diagram depicted utilizes a proportional controller; therefore the transfer function G_c is equal to K_c , the controller gain. The controller gain can be derived as an expression of K_p , τ_p , and θ_p that correspond to critical damping of the process response. This is accomplished using the Taylor series approximation and Pade 1/1 approximation. It is assumed in both cases that the effect of the disturbance on PV can be neglected.

Taylor Series Approximation

*Taylor Series
Approximation:* $e^x = 1 + x$

$$G_c = K_c$$

$$G_p = \frac{K_p * e^{-\theta_p s}}{\tau_p s + 1}$$

For the feedback control depicted the equation for the PV is as follows:

$$PV = E(s) * G_c * G_p \quad (1-1)$$

The error, $E(s)$, is the difference between the SP and PV:

$$PV = (SP - PV) * G_c * G_p \quad (1-2)$$

From this equation solve for PV:

$$PV = \frac{SP * G_c * G_p}{1 + G_c * G_p} \quad (1-3)$$

The closed loop transfer function is as follows:

$$\frac{PV}{SP} = \frac{G_c * G_p}{1 + G_c * G_p} \quad (1-4)$$

Substitute values for G_c & G_p :

$$\frac{PV}{SP} = \frac{K_c * \frac{K_p * e^{-\theta_p s}}{\tau_p s + 1}}{1 + K_c * \frac{K_p * e^{-\theta_p s}}{\tau_p s + 1}} \quad (1-5)$$

Multiply by $\tau_p s + 1$ in the numerator and denominator:

$$\frac{PV}{SP} = \frac{K_c * K_p * e^{-\theta_p s}}{\tau_p s + 1 + K_c * K_p * e^{-\theta_p s}} \quad (1-6)$$

Move the exponential term in the numerator to the denominator:

$$\frac{PV}{SP} = \frac{K_C * K_P}{[\tau_P s + 1 + K_C * K_P * e^{-\theta_P s}] * e^{\theta_P s}} \quad (1-7)$$

Substitute the exponential terms using the Taylor Approximation:

$$\frac{PV}{SP} = \frac{K_C * K_P}{[\tau_P s + 1 + K_C * K_P * (1 - \theta_P s)] * (1 + \theta_P s)} \quad (1-8)$$

The denominator is the characteristic closed loop transfer function and is set equal to 0:

$$0 = [\tau_P s + 1 + K_C * K_P * (1 - \theta_P s)] * (1 + \theta_P s) \quad (1-9)$$

The terms are multiplied out:

$$0 = \tau_P s + \tau_P \theta_P s^2 + 1 + \theta_P s + K_C K_P - K_C K_P \theta_P^2 s^2 \quad (1-10)$$

Rearrange the terms:

$$0 = (\tau_P \theta_P - K_C K_P \theta_P^2) s^2 + (\tau_P + \theta_P) s + (1 + K_C K_P) \quad (1-11)$$

Divide by $1 + K_C K_P$ in order to produce an equation similar to a second order transfer function:

$$0 = \frac{(\tau_P \theta_P - K_C K_P \theta_P^2)}{(1 + K_C K_P)} s^2 + \frac{(\tau_P + \theta_P)}{(1 + K_C K_P)} s + 1 \quad (1-12)$$

Based on a second order transfer function use the terms preceding s^2 and s to determine K_C :

$$\tau^2 = \frac{(\tau_P \theta_P - K_C K_P \theta_P^2)}{(1 + K_C K_P)} \quad (1-13a)$$

$$\tau = \sqrt{\frac{(\tau_P \theta_P - K_C K_P \theta_P^2)}{(1 + K_C K_P)}} \quad (1-13b)$$

$$2\zeta\tau = \frac{(\tau_P + \theta_P)}{(1 + K_C K_P)} \quad (\zeta = 1) \quad (1-14)$$

Substitute Equation 1.13b into 1.14:

$$2\sqrt{\frac{(\tau_P \theta_P - K_C K_P \theta_P^2)}{(1 + K_C K_P)}} = \frac{(\tau_P + \theta_P)}{(1 + K_C K_P)} \quad (1-15)$$

Set the equation equal to 0:

$$0 = 2\sqrt{\frac{(\tau_P \theta_P - K_C K_P \theta_P^2)}{(1 + K_C K_P)}} - \frac{(\tau_P + \theta_P)}{(1 + K_C K_P)} \quad (1-16)$$

Values for K_P , τ_P , θ_P are substituted and the value for K_C is determined to make Equation 1-# true using Solver in Excel. The value for K_C determined was .145.

Pade 1/1 Approximation

Pade 1/1

Approximation:

$$e^{-\theta_P s} = \frac{1 - \frac{\theta_P s}{2}}{1 + \frac{\theta_P s}{2}}$$

$$G_C = K_C$$

$$G_P = \frac{K_P * e^{-\theta_P s}}{\tau_P s + 1}$$

Following the same steps as for the Taylor Series Approximation, Equation 1-6 is obtained:

$$\frac{PV}{SP} = \frac{K_C * K_P * e^{-\theta_P s}}{\tau_P s + 1 + K_C * K_P * e^{-\theta_P s}} \quad (1-6)$$

Substitute the Pade Approximation for the exponential terms:

$$\frac{PV}{SP} = \frac{K_C * K_P * \frac{1 - \frac{\theta_P s}{2}}{1 + \frac{\theta_P s}{2}}}{\tau_P s + 1 + K_C * K_P * \frac{1 - \frac{\theta_P s}{2}}{1 + \frac{\theta_P s}{2}}} \quad (2-1)$$

Multiply the numerator and denominator by $1 + \frac{\theta_P s}{2}$:

$$\frac{PV}{SP} = \frac{K_C * K_P * 1 - \frac{\theta_P s}{2}}{\tau_P s + \frac{\theta_P \tau_P s^2}{2} + 1 + \frac{\theta_P s}{2} + K_C * K_P * (1 - \frac{\theta_P s}{2})} \quad (2-2)$$

The denominator is the characteristic closed loop transfer function and is set equal to 0:

$$0 = \tau_P s + \frac{\theta_P \tau_P s^2}{2} + 1 + \frac{\theta_P s}{2} + K_C * K_P * (1 - \frac{\theta_P s}{2}) \quad (2-3)$$

Rearrange the terms:

$$0 = \left(\frac{\tau_P \theta_P}{2}\right) s^2 + \left(\tau_P + \frac{\theta_P}{2} - \frac{K_C K_P \theta_P}{2}\right) s + (1 + K_C K_P) \quad (2-4)$$

Divide by $1 + K_C K_P$ in order to produce an equation similar to a second order transfer function:

$$0 = \frac{\left(\frac{\tau_P \theta_P}{2}\right)}{1 + K_C K_P} s^2 + \frac{\left(\tau_P + \frac{\theta_P}{2} - \frac{K_C K_P \theta_P}{2}\right)}{1 + K_C K_P} s + 1 \quad (2-5)$$

Based on a second order transfer function use the terms preceding s^2 and s to determine K_C :

$$\tau^2 = \frac{\left(\frac{\tau_P \theta_P}{2}\right)}{1 + K_C K_P} \quad (2-6a)$$

$$\tau = \sqrt{\frac{\left(\frac{\tau_P \theta_P}{2}\right)}{1 + K_C K_P}} \quad (2-6b)$$

$$2\zeta\tau = \frac{\left(\tau_P + \frac{\theta_P}{2} - \frac{K_C K_P \theta_P}{2}\right)}{1 + K_C K_P} \quad (\zeta = 1) \quad (2-7)$$

Substitute Equation 2-6b into 2-7:

$$2 \sqrt{\frac{\left(\frac{\tau_P \theta_P}{2}\right)}{1 + K_C K_P}} = \frac{\left(\tau_P + \frac{\theta_P}{2} - \frac{K_C K_P \theta_P}{2}\right)}{1 + K_C K_P} \quad (2-8)$$

Set the equation equal to 0:

$$0 = 2 \sqrt{\frac{\left(\frac{\tau_P \theta_P}{2}\right)}{1 + K_C K_P}} - \frac{\left(\tau_P + \frac{\theta_P}{2} - \frac{K_C K_P \theta_P}{2}\right)}{1 + K_C K_P} \quad (2-9)$$

Values for K_P , τ_P , θ_P are substituted and the value for K_C is determined to make Equation 1-# true using Solver in Excel. The value for K_C determined was .0972.

Problem 1.1

Using the *PIDController* file supplied, the true controller gain that corresponds to a critically damped process was determined to be .147. The errors in the derived controller gain values are:

Taylor Approximation

$$\%error = \frac{(.147 - .145)}{.147} * 100\% = 1.36\%$$

Pade 1/1 Approximation

$$\%error = \frac{(.147 - .0972)}{.147} * 100\% = 33.88\%$$

Using the estimated controller gain for the Taylor Approximation the response of the PV was:

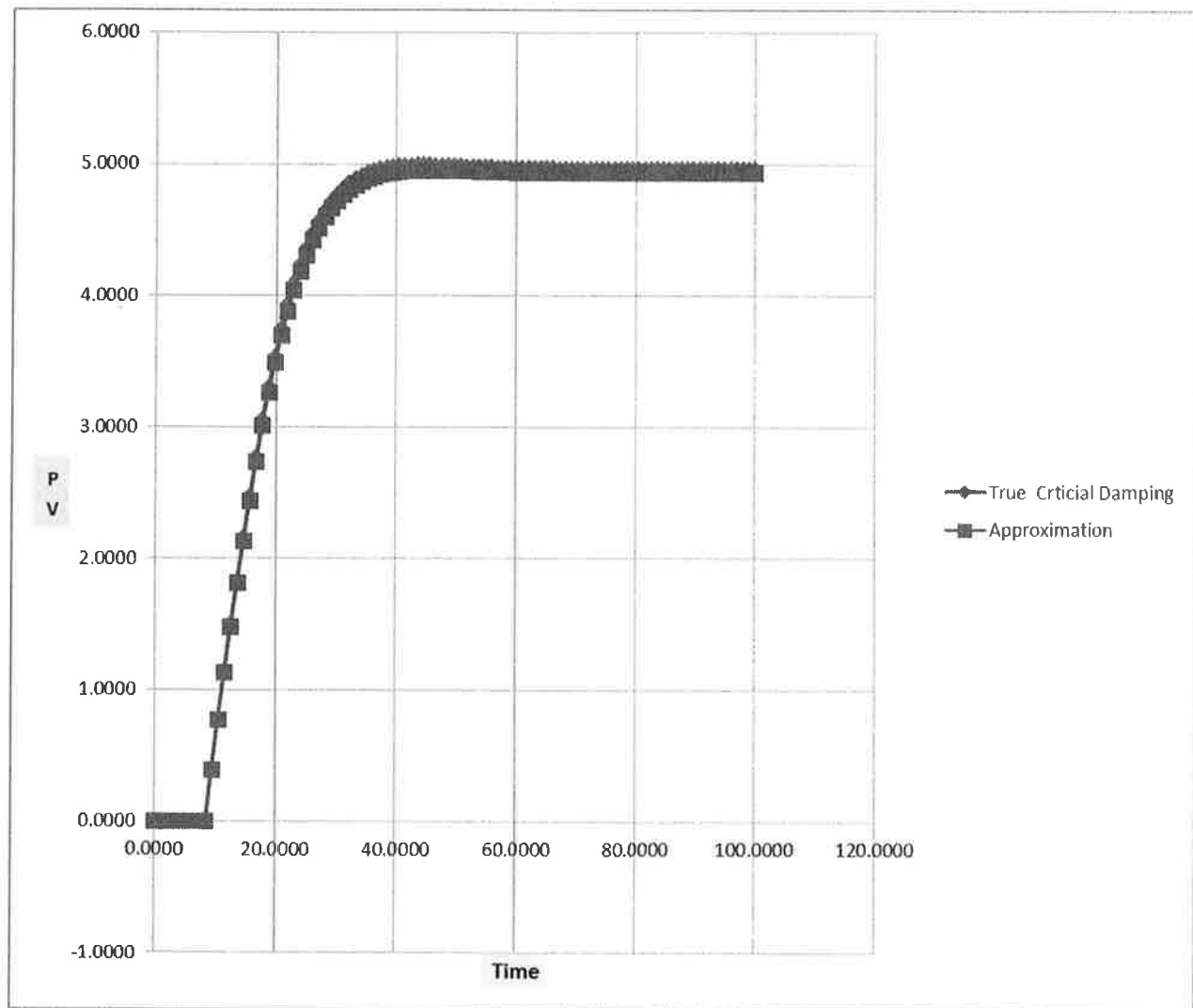


Figure 1: Process Response using Taylor Approximation Gain Controller Gain

Since the controller gain derived from the Taylor Approximation was nearly equal to the controller gain corresponding to a critically damped system, the two responses were nearly identical. It should be noted that although a set point step change of 10 was selected, the response plateaued at a value of 5. This offset is due to the use of a P controller.

Problem 1.2

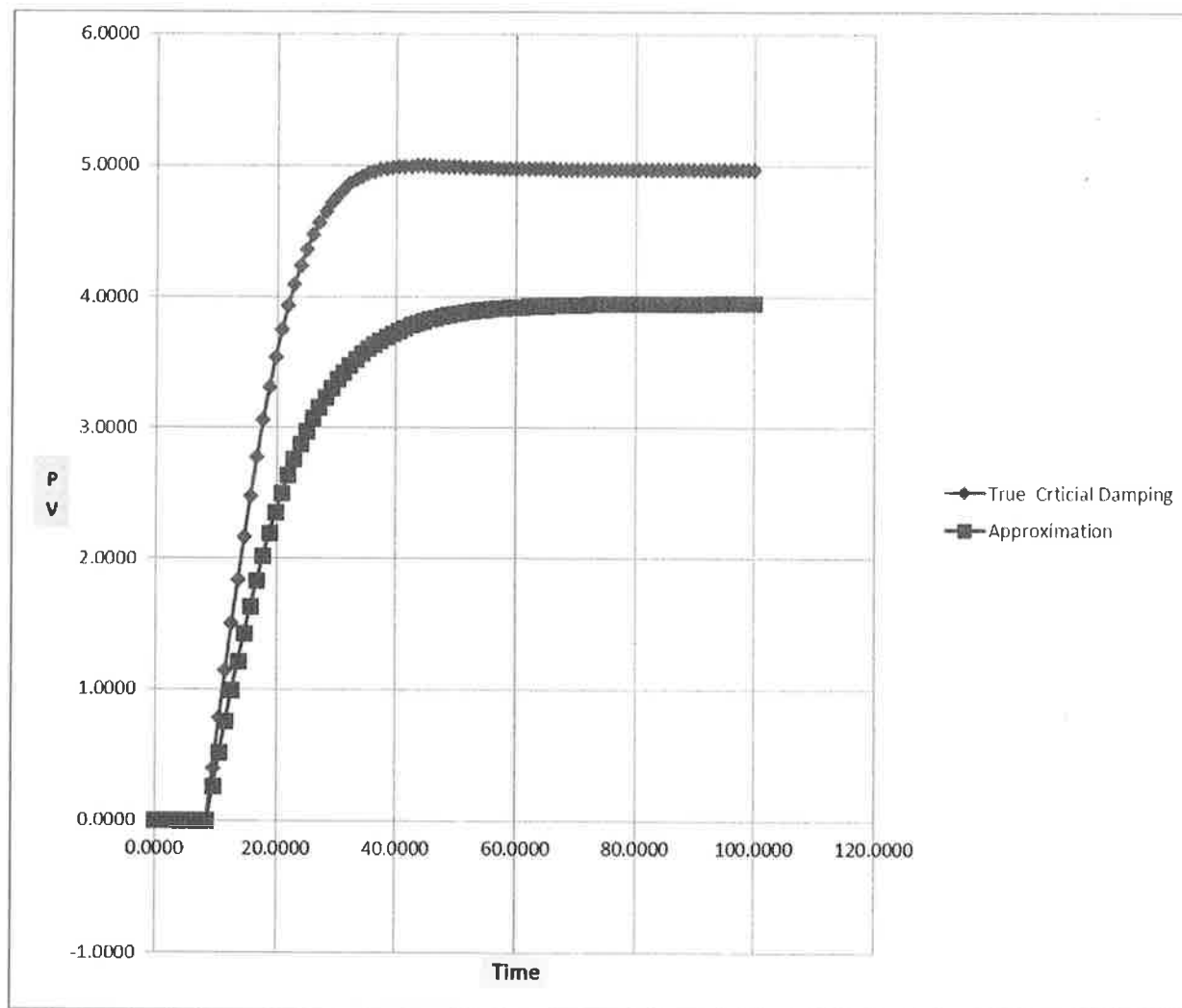


Figure 2: Process Response using Pade 1/1 Approximation Controller Gain

There was a much larger error in the estimated controller gain using the Pade 1/1 Approximation. Therefore, the PV response was observed to be significantly different from the true controller gain. Since the value for the controller gain was significantly smaller than that of the true gain, the controller was less aggressive and the response plateaued at a value of 4 with a set point change equal to 10.

Problem 1.3

Original Controller Gain from Taylor Approximation: .145

New Controller Gain (20% Increase) = $.145 * 1.2 = .174$

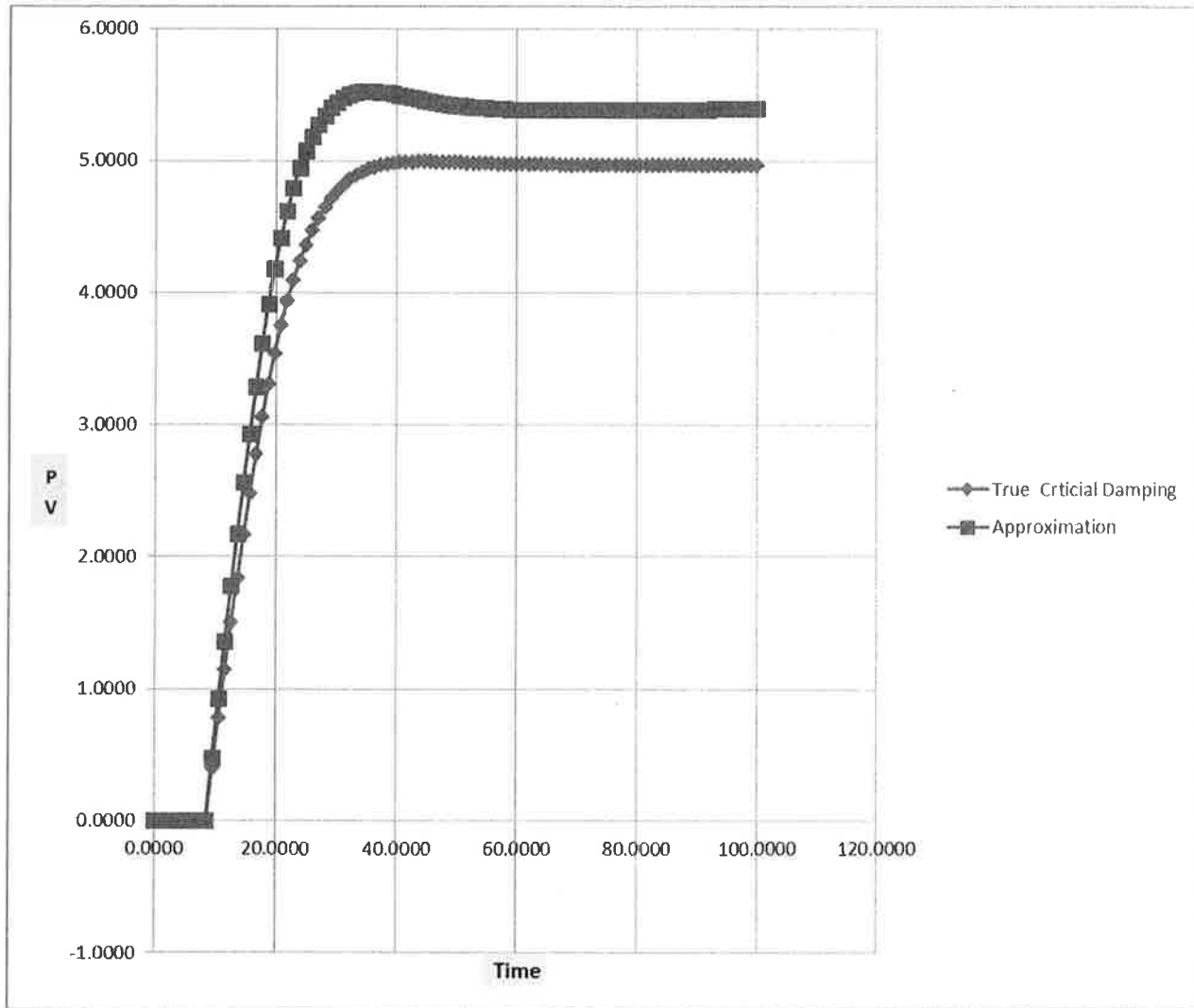


Figure 3: Process Response with 20% Increase in Taylor Approximation Controller Gain

After increasing the estimated controller gain from the Taylor Approximation by 20%, it was observed that the process response was overdamped. This is observed in Figure 1.# where the response overshoots before dropping back down and plateauing at PV=5.4. It should be noted that the process response is closer to the actual set point, SP=10, since the controller gain is even larger, thus making the controller more aggressive.

Problem 1.4

Original Controller Gain from Pade 1/1 Approximation: .0972

New Controller Gain (20% Increase) = $.0972 * .8 = .0778$

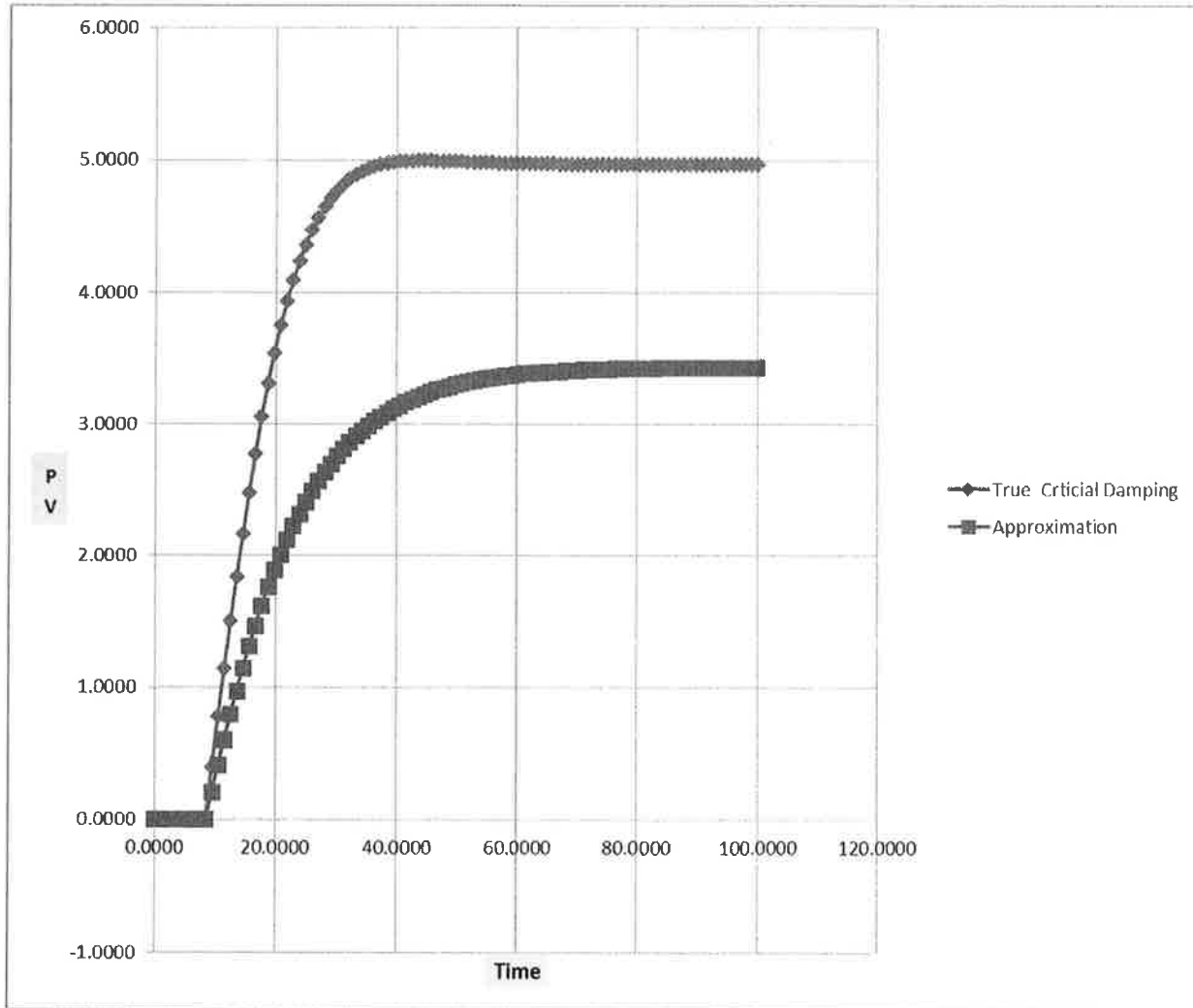


Figure 4: Process Response with 20% decrease in the Pade Approximation Controller Gain

After decreasing the estimated controller gain from the Pade 1/1 Approximation by 20%, it was observed that the process response was underdamped. This is because the process response using the approximated gain was less aggressive than the response using the true critical damping controller gain. Therefore, the process response was further from the actual set point step change equal to 10.

Problem 2.1

It is seen in the feedback control block diagram that $Y_1(s)$ is the product of $L(s)$ and G_{P11} :

$$G_{P11} = \frac{.1247}{10s + 1}$$

$$Y_1(s) = L(s) * G_{P11}$$

Substitute for G_{P11} :

$$Y_1(s) = L(s) * \frac{.1247}{10s + 1}$$

Multiply both sides by $10s+1$:

$$10s * Y_1(s) + Y_1(s) = .1247 * L(s)$$

Convert from Laplace domain to time domain:

$$10 \frac{dY_1}{dt} + Y_1(t) = .1247 * L(t)$$

Problem 2.2

It is seen in the feedback control block diagram that $Y_2(s)$ is the product of $B(s)$ and G_{P12} :

$$G_{P12} = \frac{.546}{24s + 1}$$

$$Y_2(s) = B(s) * G_{P12}$$

Substitute for G_{P12} :

$$Y_2(s) = B(s) * \frac{.546}{24s + 1}$$

Multiply both sides by $24s+1$:

$$24s * Y_2(s) + Y_2(s) = .546 * B(s)$$

Convert from Laplace domain to time domain:

$$24 \frac{dY_2}{dt} + Y_2(t) = .546 * B(t)$$

Problem 2.3

It is seen in the feedback control block diagram that $Y_3(s)$ is the product of $L(s)$ and G_{P21} :

$$G_{P21} = \frac{-.1246}{16s + 1}$$

$$Y_3(s) = L(s) * G_{P21}$$

Substitute for G_{P21} :

$$Y_3(s) = L(s) * \frac{-.1246}{16s + 1}$$

Multiply both sides by $16s+1$:

$$16s * Y_3(s) + Y_3(s) = -.1246 * L(s)$$

Convert from Laplace domain to time domain:

$$16 \frac{dY_3}{dt} + Y_3(t) = -.1246 * L(t)$$

Problem 2.4

It is seen in the feedback control block diagram that $Y_4(s)$ is equal to the product of $B(s)$ and G_{P22} :

$$G_{P22} = \frac{1.116}{22s + 1}$$

$$Y_4(s) = B(s) * G_{P22}$$

Substitute for G_{P22} :

$$Y_4(s) = B(s) * \frac{1.116}{22s + 1}$$

Multiply both sides by $22s+1$:

$$22s * Y_4(s) + Y_4(s) = 1.116 * B(s)$$

Convert from Laplace domain to time domain:

$$22 \frac{dY_4}{dt} + Y_4(t) = 1.116 * B(t)$$

Problem 2.5

It is seen in the feedback control block diagram that $PV_1(s)$ is equal to the sum of $Y_1(s)$, $Y_2(s)$, and $Y_5(s)$:

$$PV_1(s) = Y_1(s) + Y_2(s) + Y_5(s)$$

It is assumed that $Y_5(s)$, representing the disturbance, can be neglected. Therefore, the equation simplifies to:

$$PV_1(s) = Y_1(s) + Y_2(s)$$

Convert from Laplace domain to time domain:

$$PV_1(t) = Y_1(t) + Y_2(t)$$

Substitute values for $Y_1(t)$ and $Y_2(t)$ from Problems 2.1 and 2.2:

$$PV_1(t) = .1247 * L(t) - 10 \frac{dY_1}{dt} + .546 * B(t) - 24 \frac{dY_2}{dt}$$

Problem 2.6

It is seen in the feedback control block diagram that $PV_2(s)$ is equal to the sum of $Y_3(s)$, $Y_4(s)$, and $Y_6(s)$:

$$PV_2(s) = Y_3(s) + Y_4(s) + Y_6(s)$$

It is assumed that $Y_6(s)$ can be neglected, therefore the equation simplifies to:

$$PV_2(s) = Y_3(s) + Y_4(s)$$

Convert from Laplace domain to time domain:

$$PV_2(t) = Y_3(t) + Y_4(t)$$

Substitute values for $Y_1(t)$ and $Y_2(t)$ from Problems 2.1 and 2.2:

$$PV_2(t) = -.1246 * L(t) - 16 \frac{dY_3}{dt} + 1.116 * B(t) - 22 \frac{dY_4}{dt}$$

Problem 2.7*Controller 1:*

Considering only the simple feedback control loop for the distillate section of the column, $PV_1(s)$ is defined as:

$$PV_1(s) = Y_1(s) + Y_2(s) + Y_5(s)$$

It is assumed that $Y_5(s)$, representing the disturbance, can be neglected. Therefore, the equation simplifies to:

$$PV_1(s) = Y_1(s) + Y_2(s)$$

Substitute for $Y_1(s)$ and $Y_2(s)$:

$$Y_1(s) = E_1(s) * G_{C1} * G_{P11}$$

$$Y_2(s) = E_2(s) * G_{C2} * G_{P12}$$

$$PV_1(s) = E_1(s) * G_{C1} * G_{P11} + E_2(s) * G_{C2} * G_{P12}$$

From the block diagram it is seen that $E_1(s)$ is equal to the difference between SP_{XD} and $PV_1(s)$. Substitute for $E_1(s)$:

$$PV_1(s) = (SP_{XD} - PV_1(s)) * G_{C1} * G_{P11} + E_2(s) * G_{C2} * G_{P12}$$

Solve for $PV_1(s)$:

$$PV_1(s) = \frac{SP_{XD} * G_{C1} * G_{P11} + E_2(s) * G_{C2} * G_{P12}}{1 + G_{C1} * G_{P11}}$$

The denominator is the characteristic equation for a closed loop transfer function. Set this equal to 0:

$$0 = 1 + G_{C1} * G_{P11}$$

Substitute for G_{C1} and G_{P11} . A PI controller is used in this case, therefore:

$$G_{C1} = K_{C1} * \left(1 + \frac{1}{\tau_{I1}s}\right)$$

$$0 = 1 + K_{C1} * \left(1 + \frac{1}{\tau_{I1}s}\right) * \frac{.1247}{10s + 1}$$

$$0 = 1 + \frac{.1247 * K_{C1}}{10s + 1} + \frac{.1247 * K_{C1}}{\tau_{I1}s(10s + 1)}$$

Multiply both sides by $\tau_{I1}s(10s + 1)$:

$$0 = \tau_{I1}s(10s + 1) + .1247 * K_{C1} * \tau_{I1}s + .1247 * K_{C1}$$

Rearrange terms:

$$0 = (10\tau_{I1})s^2 + (\tau_{I1} + .1247K_{C1}\tau_{I1})s + .1247K_{C1}$$

Using Routh's criterion, set the term preceding s to a value greater than 0:

$$\tau_{I1}(1 + .1247K_{C1}) > 0$$

Divide by $(1 + .1247K_{C1})$ on both sides to determine the range for the integral time constant:

$$\tau_{I1} > 0$$

Controller 2:

Considering only the simple feedback control loop for the distillate section of the column, $PV_2(s)$ is defined as:

$$PV_2(s) = Y_3(s) + Y_4(s) + Y_6(s)$$

It is assumed that $Y_6(s)$, representing the disturbance, can be neglected. Therefore, the equation simplifies to:

$$PV_2(s) = Y_3(s) + Y_4(s)$$

Substitute for $Y_3(s)$ and $Y_4(s)$:

$$Y_3(s) = E_1(s) * G_{C1} * G_{P21}$$

$$Y_4(s) = E_2(s) * G_{C2} * G_{P22}$$

$$PV_2(s) = E_1(s) * G_{C1} * G_{P21} + E_2(s) * G_{C2} * G_{P22}$$

From the block diagram it is seen that $E_2(s)$ is equal to the difference between SP_{XB} and $PV_2(s)$. Substitute for $E_2(s)$:

$$PV_2(s) = E_1(s) * G_{C1} * G_{P21} + (SP_{XB} - PV_2(s)) * G_{C2} * G_{P22}$$

Solve for $PV_2(s)$:

$$PV_2(s) = \frac{E_1(s) * G_{C1} * G_{P21} + SP_{XB} * G_{C2} * G_{P22}}{1 + G_{C2} * G_{P22}}$$

The denominator is the characteristic equation for a closed loop transfer function. Set this equal to 0:

$$0 = 1 + G_{C2} * G_{P22}$$

Substitute for G_{C2} and G_{P22} . A PI controller is used in this case, therefore:

$$G_{C2} = K_{C2} * \left(1 + \frac{1}{\tau_{I2}s}\right)$$

$$0 = 1 + K_{C2} * \left(1 + \frac{1}{\tau_{I2}s}\right) * \frac{1.116}{22s + 1}$$

$$0 = 1 + \frac{1.116 * K_{C2}}{22s + 1} + \frac{1.116 * K_{C2}}{\tau_{I2}s(22s + 1)}$$

Multiply both sides by $\tau_{I2}s(22s + 1)$:

$$0 = \tau_{I2}s(22s + 1) + 1.116 * K_{C2} * \tau_{I2}s + 1.116 * K_{C2}$$

Rearrange terms:

$$0 = (22\tau_{I2})s^2 + (\tau_{I2} + 1.116K_{C2}\tau_{I2})s + 1.116K_{C2}$$

Using Routh's criterion, set the term preceding s to a value greater than 0:

$$\tau_{I2}(1 + 1.116K_{C2}) > 0$$

Divide by $(1 + 1.116K_{C2})$ on both sides to determine the range for the integral time constant:

$$\tau_{I2} > 0$$

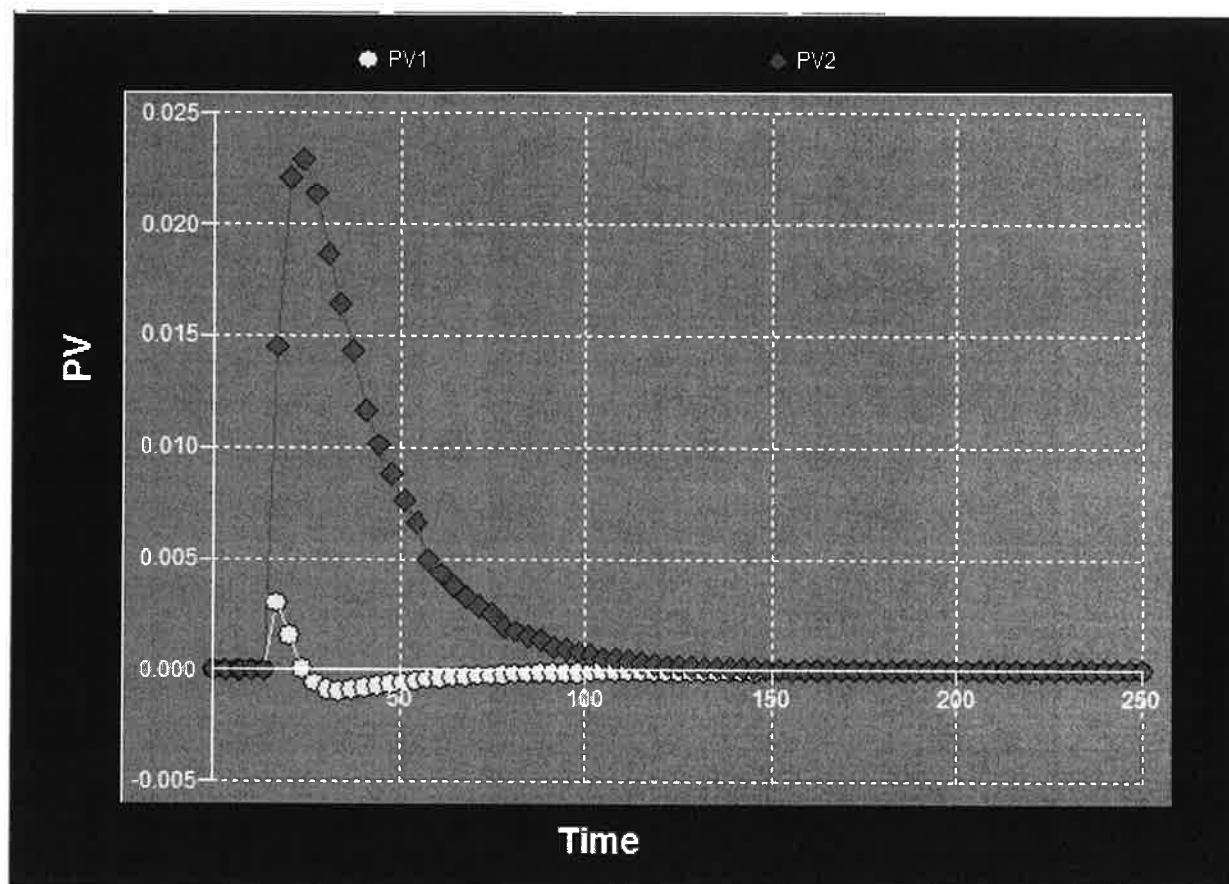
Problem 2.8

Figure 5: Process Response of Distillation Column with Set Point Change in Feed

When a step change of .1 in the feed composition was introduced into the system the two controllers were successfully able to return the process to the original set point. The controller gains were set to the given values, while the integral time constants were estimated to be equal to the process lags of 10 and 22 for controllers 1 and 2 respectively. The process is stable, however, tuning of the second controller was considered in order to get a less drastic response. By decreasing the initial integral time threefold, the following response was given.

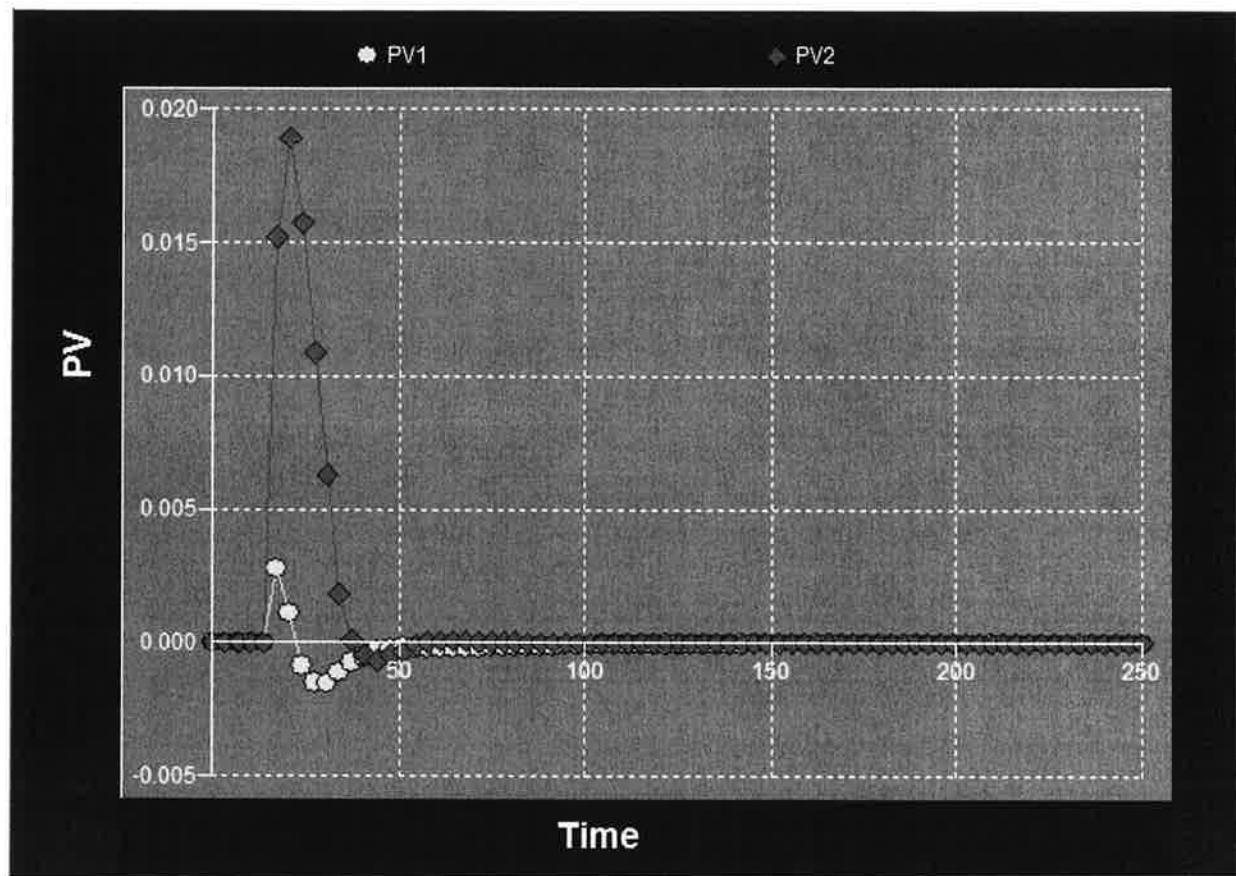


Figure 6: Process Response of Distillation Column with Set Point Change in Feed after Tuning

By decreasing the integral time, the response of PV2 decreased. Additionally, the amount of time it took for the controllers to respond and return the process variables back to their respective set points was reduced for both responses.

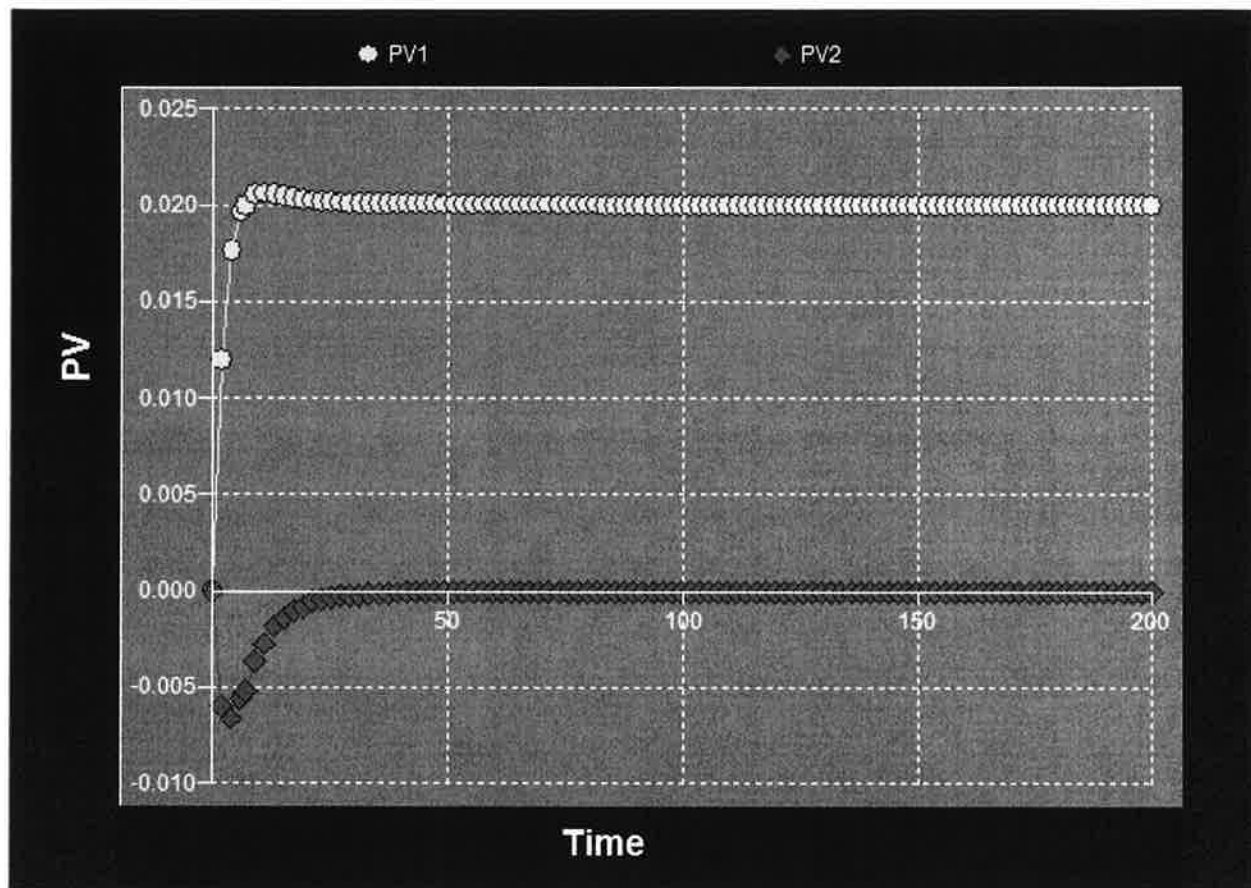
Problem 2.9

Figure 7: Process Response of Distillation Column with Set Point Change in Distillate Composition

When a step change of .02 was applied to the distillate set point, the controllers were able to successfully reach the new set point and maintain the bottoms set point. The given controller gains were used, and the integral times for each controller were once again assumed to be the corresponding process lags of 10 and 22 for controllers 1 and 2 respectively. There were very little oscillations, signifying a great amount of robustness. Additionally, the controllers were able to respond to the set point change relatively quickly. Thus, it could be considered that the controllers' performance was acceptable and no further tuning was required. However, several tests were performed to investigate if the controllers could be further tuned.

It was determined that the response of the second process variable was generally constant. However, increasing the integral time to 15 for controller 1 produced a critically damped response. This response slightly reduced the time required to reach the desired set point.

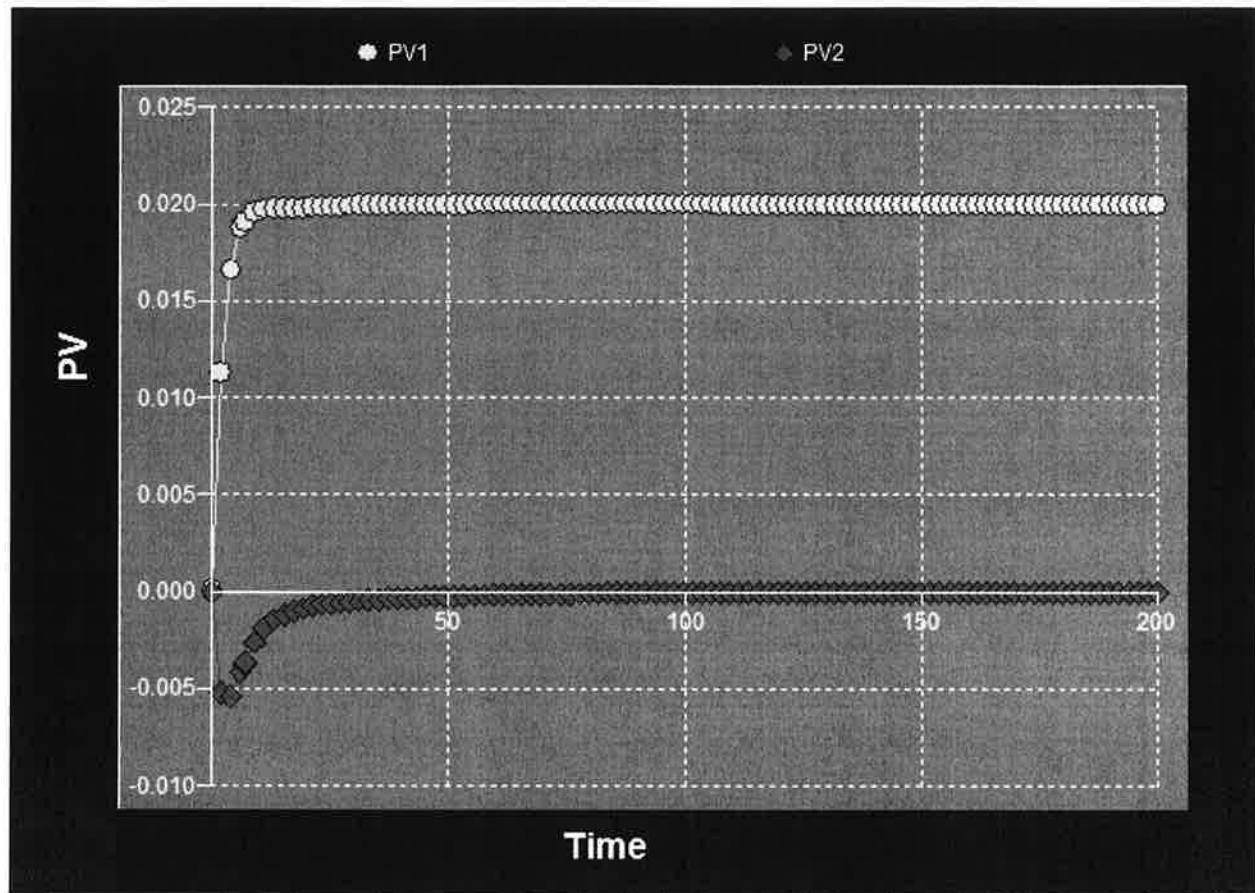


Figure 8: Process Response of Distillation Column with Set Point Change in Distillate Composition after Tuning

Problem 2.10: Proofs

In this section, a dead time of 5 minutes will be included in the composition analyzers. This will have to be accounted for in the process transfer function. To do this, a new block for the measurement element must be added on the returning PV lines. This block must have its own transfer function accounting for the time delay. Using this simple feedback control block diagram with a transportation delay in the measurement element, the equation for PV would now become:

$$PV_1 = E_1(s) * G_{C1} * G_{P11} * G_{M1}$$

Substitute for $E_1(s)$:

$$PV_1 = (SP_{XD} - PV_1) * G_{C1} * G_{P11} * G_{M1}$$

Solve for PV_1 :

$$PV_1 = \frac{SP_{XD} * G_{C1} * G_{P11} * G_{M1}}{1 + G_{C1} * G_{P11} * G_{M1}}$$

The denominator is the characteristic closed loop transfer function equation and is set equal to 0:

$$0 = 1 + G_{C1} * G_{P11} * G_{M1}$$

Substitute in the time delay for the sensor:

$$0 = 1 + G_{C1} * G_{P11} * e^{-5s}$$

This should be compared to the equation obtained prior to adding a block for the measuring device:

$$0 = 1 + G_{C1} * G_{P11}$$

Thus, it can be seen that the process transfer function G_{P11} can be used to account for both the process response and analyzer time delay.

$$G_{P11} = G_{P11} * e^{-5s}$$

This proof is applicable in for both cases in which a disturbance acts on the controller output and the process variable. When a disturbance acts on the controller output, the equation for PV is as follows:

$$PV_1 = [(SP_{XD} - PV_1) * G_{C1} + D * G_{D1}] * G_{P11} * G_{M1}$$

Solve for PV_1 :

$$PV_1 = \frac{SP_{XD} * G_{C1} * G_{P11} * G_{M1} + D * G_{D1} * G_{P11} * G_{M1}}{1 + G_{C1} * G_{P11} * G_{M1}}$$

The characteristic closed loop equation in the denominator is the same as before. When a disturbance acts on the process variable the equation for PV is as follows:

$$PV_1 = [(SP_{XD} - PV_1) * G_{C1} * G_{P11} + D * G_{D1}] * G_{M1}$$

Solve for PV_1 :

$$PV_1 = \frac{SP_{XD} * G_{C1} * G_{P11} * G_{M1} + D * G_{D1} * G_{M1}}{1 + G_{C1} * G_{P11} * G_{M1}}$$

Once again, the characteristic closed loop transfer function equation in the denominator is the same as before. Therefore, this proof is applicable to cases in which the disturbance acts on the controller output or on the process variable.

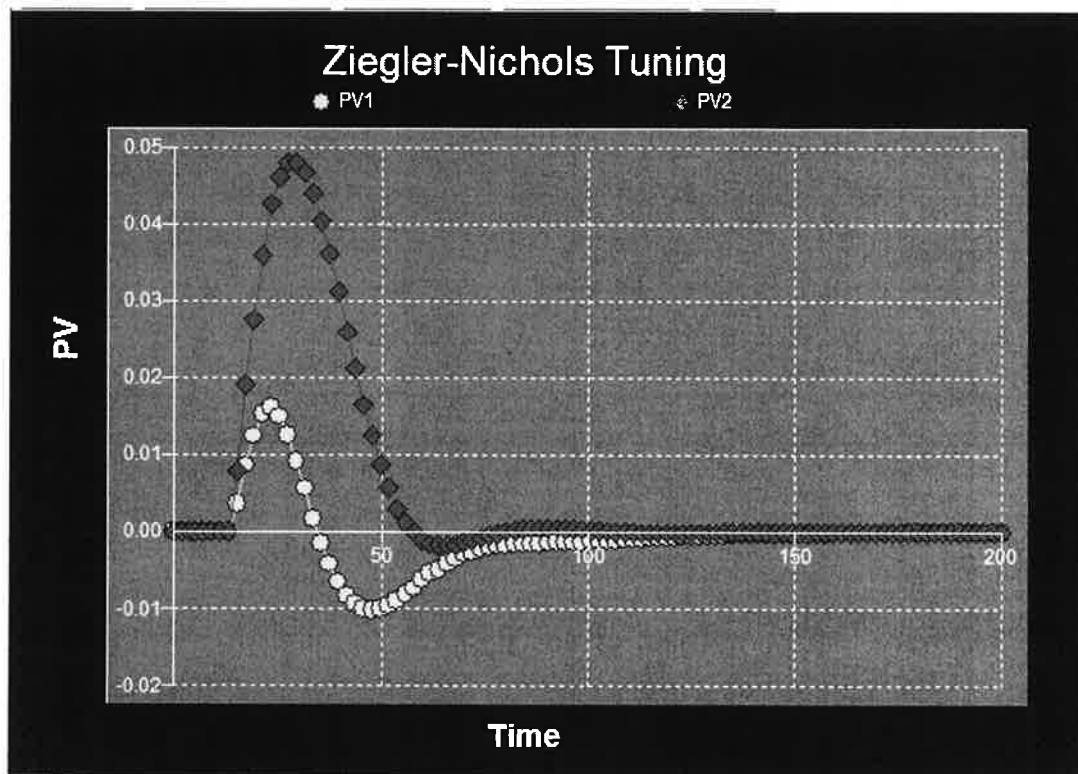
Problem 2.10-8: Regulatory Control Case

Figure 9: Process Response of Distillation Column with Set Point Change of Feed using Ziegler-Nichols Tuning

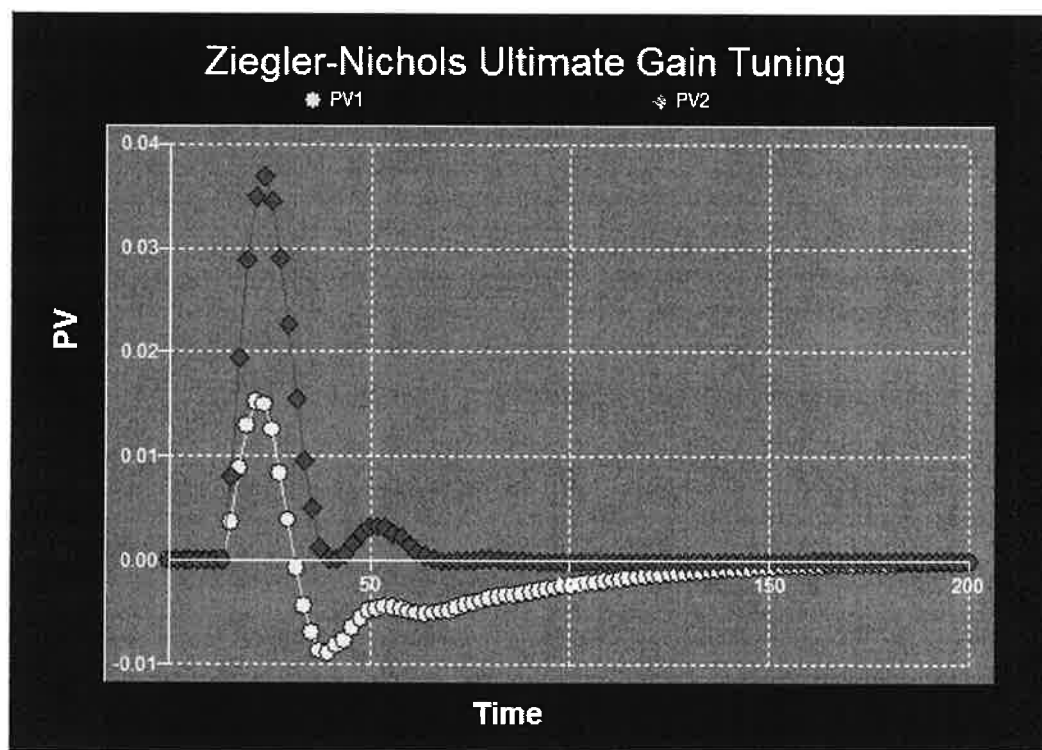


Figure 10: Process Response of Distillation Column with Set Point Change of Feed using Ziegler-Nichols Ultimate Gain Tuning

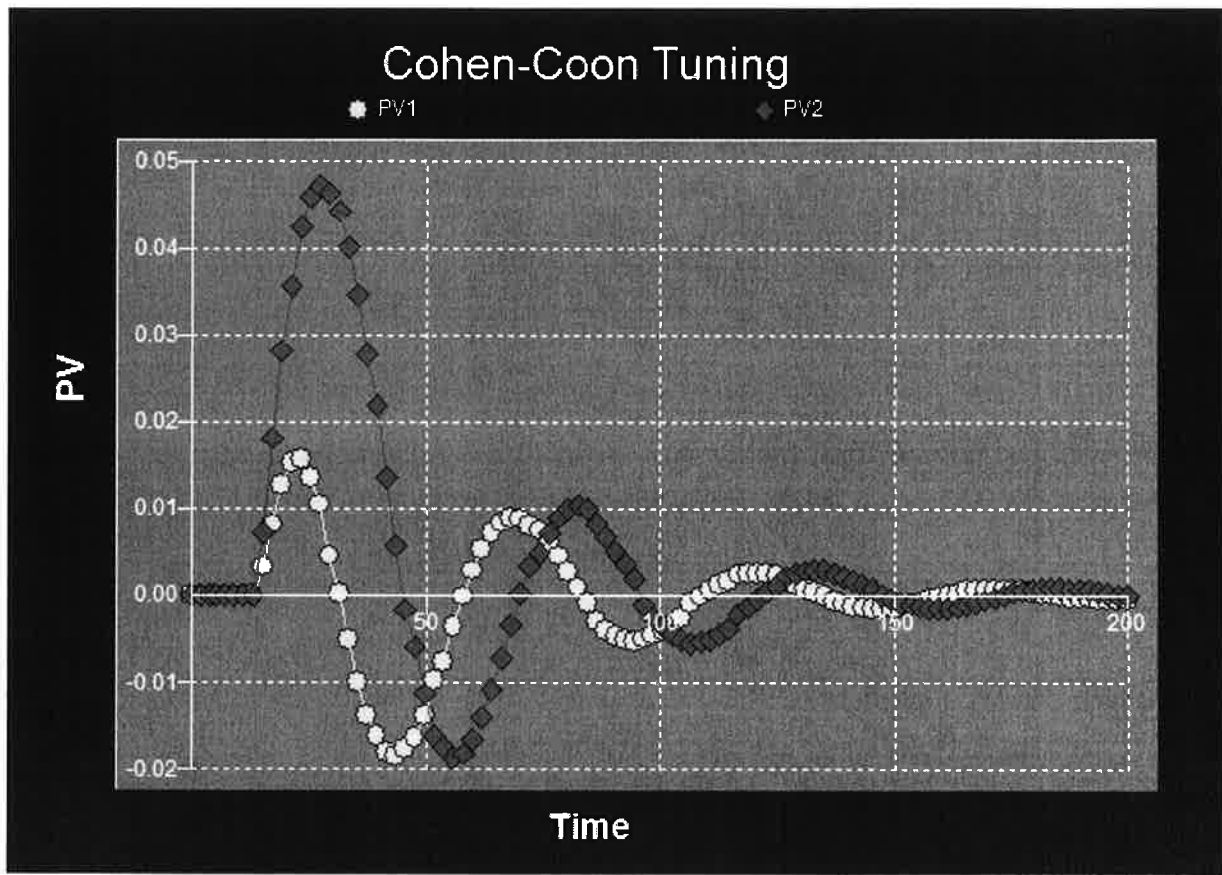


Figure 11: Process Response of Distillation Column with Set Point Change in Feed using Cohen-Coon Tuning

Table 1: Calculated Process Gains & Integral Times for Various Tuning Methods

Method	Z-N	Z-N Ultimate Gain	Cohen-Coon
Proportional Gain Controller 1	7.22	3.375	7.55
Integral Time Controller 1	16.7	15.18	8.25
Proportional Gain Controller 2	1.77	3.375	1.81
Integral Time Controller 2	16.7	15.18	11.3

Three different tuning methods were used to control the distillation column. A disturbance set point change in the feed of .1 was applied. All methods used proved to be stable. From the figures above, it can be observed that the Ziegler-Nichols tuning method, shown in Figure 9, proved to be most successful of the methods. This is because it was the quickest of all the methods to return the distillate and bottoms to their original set points while still producing a relatively robust response. Ziegler-Nichols Ultimate Gain tuning was similarly robust but took

somewhat longer to reach the set points, while Cohen-Coon was least effective since the oscillations continued for the longest period of time, proving to lack both speed and robustness.

Problem 2.10-9: Servo Control Case

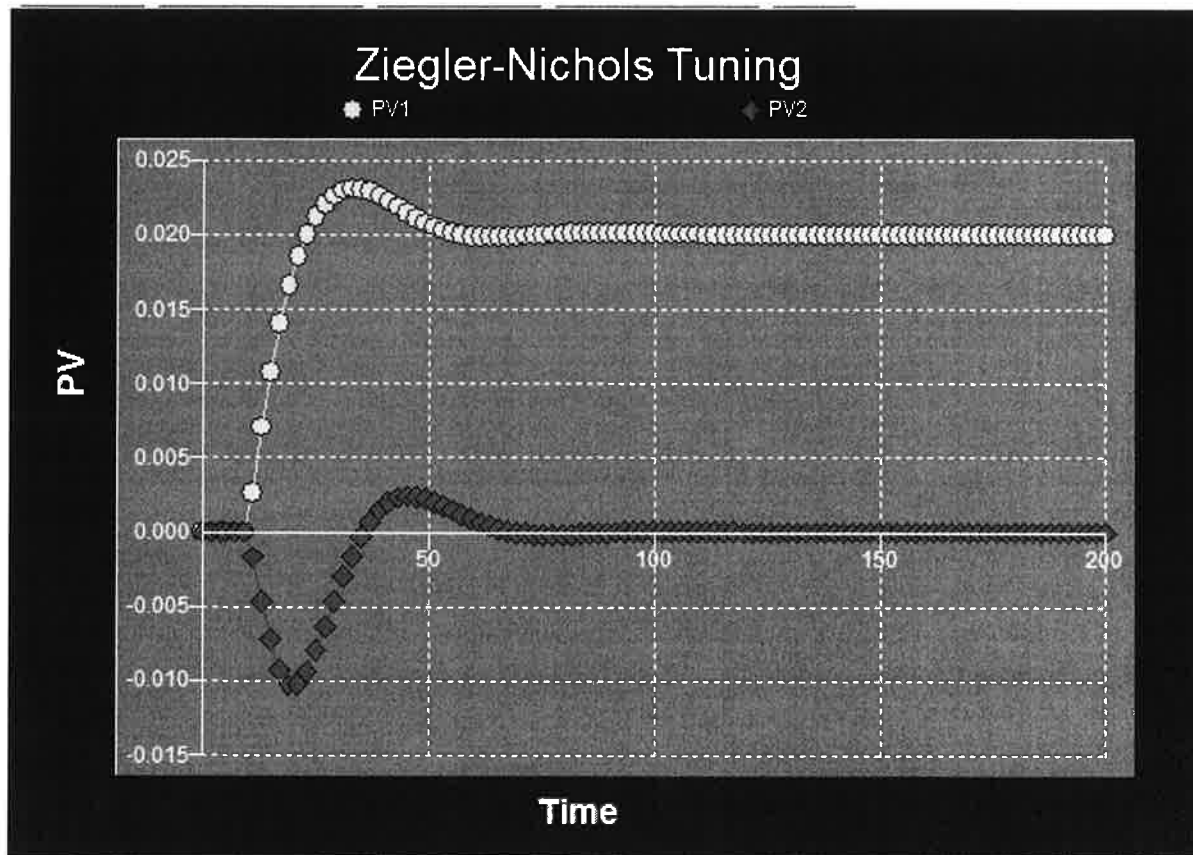


Figure 12: Process Response of Distillation Column with Set Point Change in Distillate Composition using Ziegler-Nichols Tuning

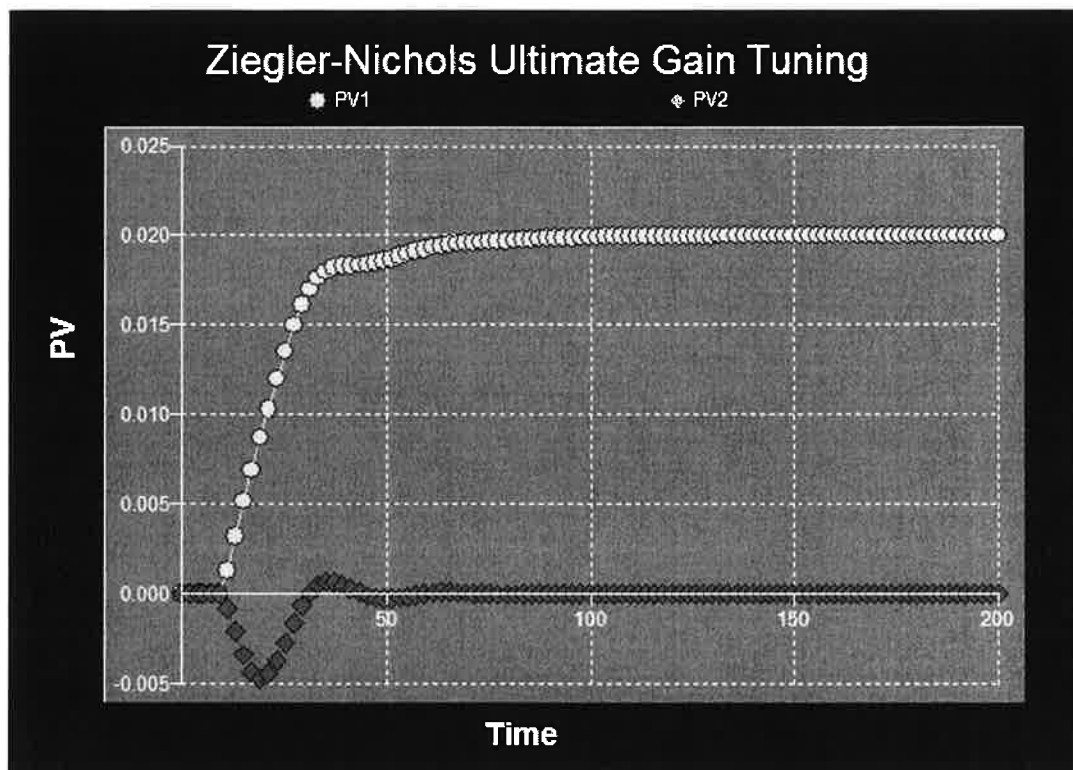


Figure 13: Process Response of Distillation Column with Set Point Change in Distillate Composition using Ziegler-Nichols Ultimate Gain Tuning

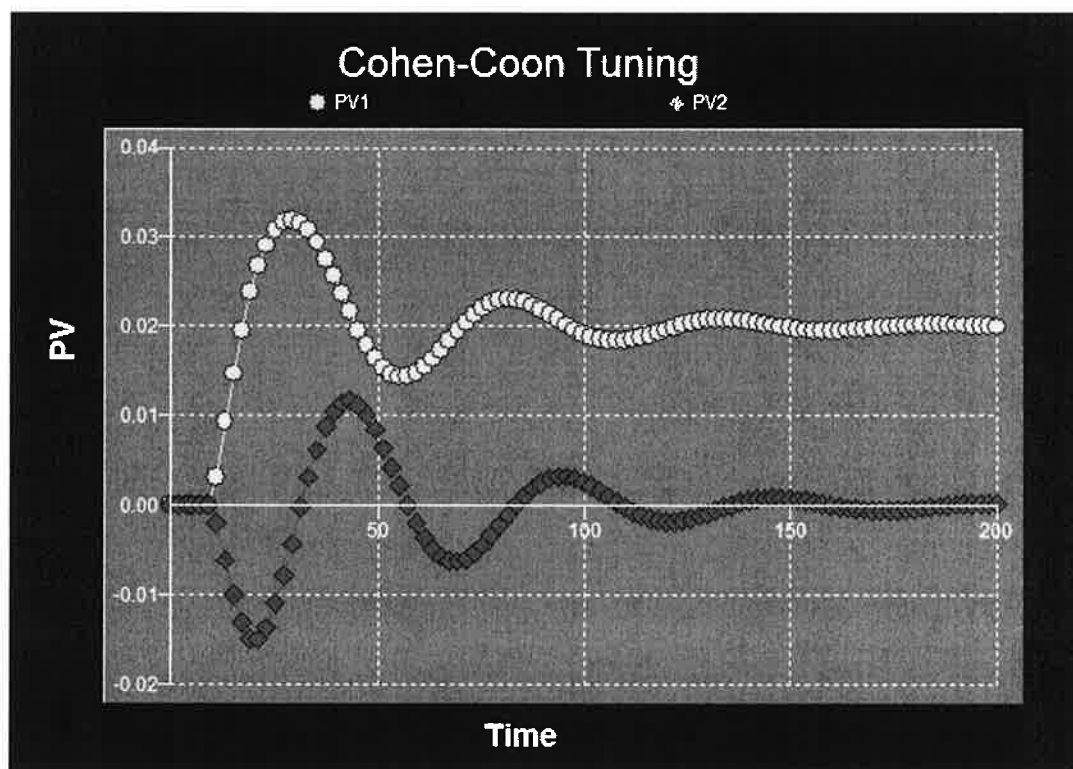


Figure 14: Process Response of Distillate Column with Set Point Change in Distillate Composition using Cohen-Coon Tuning

Table 2: Calculated Process Gains & Integral Times for Various Tuning Methods

Method	Z-N	Z-N Ultimate Gain	Cohen-Coon
Proportional Gain Controller 1	7.22	3.375	7.55
Integral Time Controller 1	16.7	15.015	8.25
Proportional Gain Controller 2	1.77	3.375	1.81
Integral Time Controller 2	16.7	15.015	11.3

A set point change in the distillate of .02 was implemented. From the three methods used, it seems that the Ziegler-Nichols Ultimate Gain method shown in Figure 13 was most effective. This is due to its relative speed and robustness compared to the other methods. Although the Ziegler-Nichols method was just as quick, the oscillations in the process variables were larger than those of the Ziegler-Nichols Ultimate Gain method, making it less robust. Additionally, this process looked to be underdamped while the Ziegler-Nichols Ultimate Gain was overdamped. The Cohen-Coon tuning method once again proved unfavorable since the response was stable but had continued oscillations and took a longer period of time for the process variables to reach the desired set points.

```

!PROBLEM 2.8 - Feed Set Point Change without Analyzer Time Delay
!PID Controllers
!-----
!Regulatory-Control Implementation in Time-Domain
!-----
!Written by: Professor Michael Caracotsios
!*****
Use AthenaUtilities
Global
KP11,KP12,KD1,KP21,KP22,KD2,Tau11,Tau12,TauD1,Tau21,Tau22,TauD2,Kc1,Ki1,Kc2,Ki2,SPxd,SPxb,DSP,PV
1,PV2,B,L As Real
Global Time(2000),Y1(2000),Y2(2000),Y3(2000),Y4(2000),Y5(2000),Y6(2000),E1(2000),E2(2000) As
Real
Global iCount As Integer

KP11=0.1247      !ND,Process Gain for Transfer Function Gp11
KP12=0.546       !ND,Process Gain for Transfer Function Gp12
KD1=0.588        !ND,Process Gain for Transfer Function Gd1
KP21=-0.1246     !ND,Process Gain for Transfer Function Gp21
KP22=1.116       !ND,Process Gain for Transfer Function Gp22
KD2=1.41         !ND,Process Gain for Transfer Function Gd2

Tau11=10         !ND,Process Lag for Transfer Function Gp11
Tau12=24         !ND,Process Lag for Transfer Function Gp12
TauD1=22         !ND,Process Lag for Transfer Function Gd1
Tau21=16         !ND,Process Lag for Transfer Function Gp21
Tau22=22         !ND,Process Lag for Transfer Function Gp22
TauD2=24         !ND,Process Lag for Transfer Function Gd2

Kc1=32.08        !ND,Proportional Gain for Controller 1
Ki1=32.08/10     !ND,Integral Gain for Controller 1

Kc2=3.58         !ND,Proportional Gain for Controller 2
Ki2=3.58/22     !ND,Integral Gain for Controller 2

iCount=0         !ND,Counter for History Profiles
SPxd=0.0         !ND,Distillate Set Point Value
SPxb=0           !ND,Bottoms Set Point Value
DSP=0.1         !ND,Disturbance Set Point Value

@Initial Conditions
U(1)=0           !ND,Initial Conditions of Y's
U(2)=0
U(3)=0
U(4)=0
U(5)=0
U(6)=0

@Model Equations
Dim IE1,IE2 As Real
Dim i As Integer

!Collect the History of Integrated Model Variables
!-----
If (iRes==9) Then
  iCount=iCount+1
  Time(iCount)=T
  Y1(iCount)=U(1)
  Y2(iCount)=U(2)
  Y3(iCount)=U(3)
  Y4(iCount)=U(4)
  Y5(iCount)=U(5)
  Y6(iCount)=U(6)
EndIf

!Formulate the Feedback Control Loop Equations

```

```

-----
Select Case(iRange)

    Case(0)                                !ND,Before Disturbance Effects
    PV1=U(1)+U(2)                          !ND,Process Variable 1 Eqn
    PV2=U(4)+U(5)                          !ND,Process Variable 2 Eqn

    If(iRes==9)E1(iCount)=SPxd-PV1 !ND,Trapezoid Rule for Integral Action of Controller 1
    IE1=0.0
    Do i=1,iCount-1
    IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

    L=Kc1*(SPxd-PV1)+Ki1*IE1                !ND,Controller 1 Output

    If(iRes==9)E2(iCount)=SPxb-PV2 !ND,Trapezoid Rule for Integral Action of Controller 2
    IE2=0.0
    Do i=1,iCount-1
    IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE2=IE2+(E2(iCount)+SPxb-PV2)*(T-Time(iCount))/2.0

    B=Kc2*(SPxb-PV2)+Ki2*IE2                !ND,Controller 2 Output

    F(1)=-U(1)+KP11*L
    F(2)=-U(2)+KP12*B
    F(3)=-U(3)
    F(4)=-U(4)+KP21*L
    F(5)=-U(5)+KP22*B
    F(6)=-U(6)

    Case(1)                                !ND,With Disturbance Effects

    PV1=U(1)+U(2)+QuadraticSpline(Time(1:iCount),Y3(1:iCount),T-7)    !ND,Process Var 1 Eqn
    PV2=U(4)+U(5)+QuadraticSpline(Time(1:iCount),Y6(1:iCount),T-7)    !ND,Process Var 2 Eqn

    If(iRes==9)E1(iCount)=SPxd-PV1    !ND,Trapezoid Rule for Integral Action of
Controller 1
    IE1=0.0
    Do i=1,iCount-1
    IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

    L=Kc1*(SPxd-PV1)+Ki1*IE1                !ND,Controller 1 Output

    If(iRes==9)E2(iCount)=SPxb-PV2    !ND,Trapezoid Rule for Integral Action of
Controller 2
    IE2=0.0
    Do i=1,iCount-1
    IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE2=IE2+(E2(iCount)+SPxb-PV2)*(T-Time(iCount))/2.0

    B=Kc2*(SPxb-PV2)+Ki2*IE2                !ND,Controller 2 Output

    F(1)=-U(1)+KP11*L
    F(2)=-U(2)+KP12*B
    F(3)=-U(3)+KD1*DSP
    F(4)=-U(4)+KP21*L
    F(5)=-U(5)+KP22*B
    F(6)=-U(6)+KD2*DSP

End Select

```

@Coefficient Matrix

```
E(1)=Tau11          !ND,Tau's for differential eqns
E(2)=Tau12
E(3)=TauD1
E(4)=Tau21
E(5)=Tau22
E(6)=TauD2
```

@Solver Options

```
Neq=6
Npts=100
Tend=200.0
Tstop=7
```

```
Param(28)=1          !ND,Continue Integration Past Dead Time
Param(177)=1         !ND,Enable Integration History Access
VariableNames=Time;Y1;Y2;Y3;Y4;Y5;Y6 !ND,Variable Names
Append=Pv1;Pv2       !ND,Display Process Variables
```

```

!PROBLEM 2.9 - Distillate Composition Set Point Change without Analyzer Time Delay
!PID Controllers
!-----
!Servo-Control Implementation in Time-Domain
!-----
!Written by: Professor Michael Caracotsios
!*****
!Use AthenaUtilities
!Global
KP11,KP12,KD1,KP21,KP22,KD2,Tau11,Tau12,TauD1,Tau21,Tau22,TauD2,Kc1,Ki1,Kc2,Ki2,SPxd,SPxb,DSP,PV
1,PV2,B,L As Real
!Global Time(2000),Y1(2000),Y2(2000),Y3(2000),Y4(2000),Y5(2000),Y6(2000),E1(2000),E2(2000) As
Real
!Global iCount As Integer

KP11=0.1247      !ND,Process Gain for Transfer Function Gp11
KP12=0.546       !ND,Process Gain for Transfer Function Gp12
KD1=0.588        !ND,Process Gain for Transfer Function Gd1
KP21=-0.1246     !ND,Process Gain for Transfer Function Gp21
KP22=1.116       !ND,Process Gain for Transfer Function Gp22
KD2=1.41         !ND,Process Gain for Transfer Function Gd2

Tau11=10         !ND,Process Lag for Transfer Function Gp11
Tau12=24         !ND,Process Lag for Transfer Function Gp12
TauD1=22         !ND,Process Lag for Transfer Function Gd1
Tau21=16         !ND,Process Lag for Transfer Function Gp21
Tau22=22         !ND,Process Lag for Transfer Function Gp22
TauD2=24         !ND,Process Lag for Transfer Function Gd2

Kc1=32.08        !ND,Proportional Gain for Controller 1
Ki1=32.08/10     !ND,Integral Gain for Controller 1

Kc2=5            !ND,Proportional Gain for Controller 2
Ki2=3.58/22     !ND,Integral Gain for Controller 2

iCount=0         !ND,Counter for History Profiles
SPxd=0.02        !ND,Distillate Set Point Value
SPxb=0           !ND,Bottoms Set Point Value
DSP=0           !ND,Disturbance Set Point Value

@Initial Conditions
U(1)=0           !ND,Initial Conditions of Y's
U(2)=0
U(3)=0
U(4)=0
U(5)=0
U(6)=0

@Model Equations
Dim IE1,IE2 As Real
Dim i As Integer

!Collect the History of the Integrated Model Variables
!-----
If (iRes==9)Then
  iCount=iCount+1
  Time(iCount)=T
  Y1(iCount)=U(1)
  Y2(iCount)=U(2)
  Y3(iCount)=U(3)
  Y4(iCount)=U(4)
  Y5(iCount)=U(5)
  Y6(iCount)=U(6)
EndIf

!Formulate the Feedback Control Loop Equations
!-----

```

```

PV1=U(1)+U(2)
PV2=U(4)+U(5)

If(iRes==9)E1(iCount)=SPxd-PV1
Integral Action of Controller 1
IE1=0.0
Do i=1,iCount-1
IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
EndDo
IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

L=Kc1*(SPxd-PV1)+Ki1*IE1

If(iRes==9)E2(iCount)=SPxb-PV2
IE2=0.0
Do i=1,iCount-1
IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
EndDo
IE2=IE2+(E2(iCount)+SPxb-PV2)*(T-Time(iCount))/2.0

B=Kc2*(SPxb-PV2)+Ki2*IE2

F(1)=-U(1)+KP11*L
F(2)=-U(2)+KP12*B
F(3)=-U(3)
F(4)=-U(4)+KP21*L
F(5)=-U(5)+KP22*B
F(6)=-U(6)

@Coefficient Matrix
E(1)=Tau11
E(2)=Tau12
E(3)=TauD1
E(4)=Tau21
E(5)=Tau22
E(6)=TauD2

@Solver Options
Neq=6
Npts=100
Tend=200.0

Param(28)=1
Param(177)=1
VariableNames=Time;Y1;Y2;Y3;Y4;Y5;Y6
Append=PV1;PV2

!ND,Process Variable 1 Eqn
!ND,Process Variable 2 Eqn
!ND,Trapezoid Rule for
!ND,Controller 1 Output
!ND,Controller 2 Output
!ND,Tau's for differential eqns
!ND,Continue Integration Past Dead Time
!ND,Enable Integration History Access
!ND,Variable Names
!ND,Display Process Variable

```

```

!Problem 10.8 - Feed Set Point Change with Analyzer Time Delay
!PID Controllers
!-----
!Regulatory-Control Implementation in Time-Domain
!-----
!Written by: Professor Michael Caracotsios
!*****
Use AthenaUtilities
Use AthenaUtilities
Global
KP11,KP12,KD1,KP21,KP22,KD2,Tau11,Tau12,TauD1,Tau21,Tau22,TauD2,Kc1,Ki1,Kc2,Ki2,SPxd,SPxb,DSP,PV
1,PV2,B,L As Real
Global Time(2000),Y1(2000),Y2(2000),Y3(2000),Y4(2000),Y5(2000),Y6(2000),E1(2000),E2(2000) As
Real
Global iCount As Integer

KP11=0.1247      !ND,Process Gain for Transfer Function Gp11
KP12=0.546       !ND,Process Gain for Transfer Function Gp12
KD1=0.588        !ND,Process Gain for Transfer Function Gd1
KP21=-0.1246     !ND,Process Gain for Transfer Function Gp21
KP22=1.116       !ND,Process Gain for Transfer Function Gp22
KD2=1.41         !ND,Process Gain for Transfer Function Gd2

Tau11=10         !ND,Process Lag for Transfer Function Gp11
Tau12=24         !ND,Process Lag for Transfer Function Gp12
TauD1=22         !ND,Process Lag for Transfer Function Gd1
Tau21=16         !ND,Process Lag for Transfer Function Gp21
Tau22=22         !ND,Process Lag for Transfer Function Gp22
TauD2=24         !ND,Process Lag for Transfer Function Gd2

Kc1=3.375        !ND,Proportional Gain for Controller 1
Ki1=3.375/15.18  !ND,Integral Gain for Controller 1

Kc2=3.375        !ND,Proportional Gain for Controller 2
Ki2=3.375/15.18  !ND,Integral Gain for Controller 2

iCount=0         !ND,Counter for History Profiles
SPxd=0.0         !ND,Distillate Set Point Value
SPxb=0           !ND,Bottoms Set Point Value
DSP=0.1          !ND,Disturbance Set Point Value

@Initial Conditions
U(1)=0           !ND,Initial Conditions of Y's
U(2)=0
U(3)=0
U(4)=0
U(5)=0
U(6)=0

@Model Equations
Dim IE1,IE2 As Real
Dim i As Integer

!Collect the History of the Integrated Model Variables
!-----
If (iRes==9)Then
  iCount=iCount+1
  Time(iCount)=T
  Y1(iCount)=U(1)
  Y2(iCount)=U(2)
  Y3(iCount)=U(3)
  Y4(iCount)=U(4)
  Y5(iCount)=U(5)
  Y6(iCount)=U(6)
EndIf

```

```

!Formulate the Feedback Control Loop Equations
!-----
Select Case(iRange)
Case(0)
    PV1=U(1)+U(2)
    PV2=U(4)+U(5)
    !ND,Analyzer & Disturbance Dead Time
    !ND,Process Variable 1 Eqn
    !ND,Process Variable 2 Eqn

    If(iRes==0)E1(iCount)=SPxd-PV1
    L=Kc1*(SPxd-PV1)+Ki1*(SPxd-PV1)*T
    !ND,Controller 1 Output

    If(iRes==0)E2(iCount)=SPxb-PV2
    B=Kc2*(SPxb-PV2)+Ki2*(SPxb-PV2)*T
    !ND,Controller 2 Output

    F(1)=-U(1)+KP11*L
    F(2)=-U(2)+KP12*B
    F(3)=-U(3)
    F(4)=-U(4)+KP21*L
    F(5)=-U(5)+KP22*B
    F(6)=-U(6)

Case(1)
    !ND,Analyzer Dead Time Ends
    PV1=QuadraticSpline(Time(1:iCount),Y1(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y2(1:iCount),T-5)
    !Process Variable 1 Eqn
    PV2=QuadraticSpline(Time(1:iCount),Y4(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y5(1:iCount),T-5)
    !Process Variable 2 Eqn

    If(iRes==9)E1(iCount)=SPxd-PV1
    IE1=0.0
    !ND,Trapezoid Rule for Integral Action of Controller 1
    Do i=1,iCount-1
    IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

    L=Kc1*(SPxd-PV1)+Ki1*IE1
    !ND,Controller 1 Output

    If(iRes==9)E2(iCount)=SPxb-PV2
    IE2=0.0
    !ND,Trapezoid Rule for Integral Action of Controller 2
    Do i=1,iCount-1
    IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE2=IE2+(E2(iCount)+SPxd-PV2)*(T-Time(iCount))/2.0

    B=Kc2*(SPxb-PV2)+Ki2*IE2
    !ND,Controller 2 Output

    F(1)=-U(1)+KP11*L
    F(2)=-U(2)+KP12*B
    F(3)=-U(3)
    F(4)=-U(4)+KP21*L
    F(5)=-U(5)+KP22*B
    F(6)=-U(6)

Case(2)
    !ND,Disturbance Dead Time Ends

    PV1=QuadraticSpline(Time(1:iCount),Y1(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y2(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y3(1:iCount),T-7)
    PV2=QuadraticSpline(Time(1:iCount),Y4(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y5(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y6(1:iCount),T-7)

    If(iRes==9)E1(iCount)=SPxd-PV1
    IE1=0.0
    !ND,Trapezoid Rule for Integral Action of Controller 1
    Do i=1,iCount-1
    IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

```

```

L=Kc1*(SPxd-PV1)+Ki1*IE1          !ND,Controller 1 Output

If(iRes==9)E2(iCount)=SPxb-PV2      !ND,Trapezoid Rule for Integral Action of Controller 2
IE2=0.0
Do i=1,iCount-1
IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
EndDo
IE2=IE2+(E2(iCount)+SPxd-PV2)*(T-Time(iCount))/2.0

B=Kc2*(SPxb-PV2)+Ki2*IE2          !ND,Controller 2 Output

F(1)=-U(1)+KP11*L
F(2)=-U(2)+KP12*B
F(3)=-U(3)+KD1*DSP
F(4)=-U(4)+KP21*L
F(5)=-U(5)+KP22*B
F(6)=-U(6)+KD2*DSP

End Select

```

@Coefficient Matrix

```

E(1)=Tau11          !ND,Tau's for differential eqns
E(2)=Tau12
E(3)=TauD1
E(4)=Tau21
E(5)=Tau22
E(6)=TauD2

```

@Solver Options

```

Neq=6
Npts=100
Tend=200.0
Tstop=5;7

Param(28)=1          !ND,Continue Integration Past Dead Time
Param(177)=1         !ND,Enable Integration History Access
VariableNames=Time;Y1;Y2;Y3;Y4;Y5;Y6 !ND,Variable Names
Append=PV1;PV2       !ND,Display Process Variable

```

```

!Problem 10.9 - Distillate Composition Set Point Change with Analyzer Time Delay
!PID Controllers
!-----
!Servo-Control Implementation in Time-Domain
!-----
!Written by: Professor Michael Caracotsios
!*****
Use AthenaUtilities
Global
  Global
KP11,KP12,KD1,KP21,KP22,KD2,Tau11,Tau12,TauD1,Tau21,Tau22,TauD2,Kc1,Ki1,Kc2,Ki2,SPxd,SPxb,DSP,PV
1,PV2,B,L As Real
  Global Time(2000),Y1(2000),Y2(2000),Y3(2000),Y4(2000),Y5(2000),Y6(2000),E1(2000),E2(2000) As
Real
  Global iCount As Integer

  KP11=0.1247      !ND,Process Gain for Transfer Function Gp11
  KP12=0.546       !ND,Process Gain for Transfer Function Gp12
  KD1=0.588        !ND,Process Gain for Transfer Function Gd1
  KP21=-0.1246     !ND,Process Gain for Transfer Function Gp21
  KP22=1.116       !ND,Process Gain for Transfer Function Gp22
  KD2=1.41         !ND,Process Gain for Transfer Function Gd2

  Tau11=10         !ND,Process Lag for Transfer Function Gp11
  Tau12=24         !ND,Process Lag for Transfer Function Gp12
  TauD1=22         !ND,Process Lag for Transfer Function Gd1
  Tau21=16         !ND,Process Lag for Transfer Function Gp21
  Tau22=22         !ND,Process Lag for Transfer Function Gp22
  TauD2=24         !ND,Process Lag for Transfer Function Gd2

  Kc1=7.55         !ND,Proportional Gain for Controller 1
  Ki1=7.55/8.25    !ND,Integral Gain for Controller 1

  Kc2=1.81         !ND,Proportional Gain for Controller 2
  Ki2=1.81/11.3    !ND,Integral Gain for Controller 2

  iCount=0         !ND,Counter for History Profiles
  SPxd=0.02        !ND,Distillate Set Point Value
  SPxb=0           !ND,Bottoms Set Point Value
  DSP=0            !ND,Disturbance Set Point Value

@Initial Conditions
  U(1)=0           !ND,Initial Conditions of Y's
  U(2)=0
  U(3)=0
  U(4)=0
  U(5)=0
  U(6)=0

@Model Equations
  Dim IE1,IE2 As Real
  Dim i As Integer

!Collect the History of the Integrated Model Variables
!-----
If (iRes==9)Then
  iCount=iCount+1
  Time(iCount)=T
  Y1(iCount)=U(1)
  Y2(iCount)=U(2)
  Y3(iCount)=U(3)
  Y4(iCount)=U(4)
  Y5(iCount)=U(5)
  Y6(iCount)=U(6)
EndIf

!Formulate the Feedback Control Loop Equations

```

```

!-----
Select Case(iRange)
Case(0)          !ND,Analyzer Dead Time

    PV1=U(1)+U(2)          !ND,Process Variable 1 Eqn
    PV2=U(4)+U(5)          !ND,Process Variable 2 Eqn

    If(iRes==9)E1(iCount)=SPxd-PV1
    L=Kc1*(SPxd-PV1)+Ki1*(SPxd-PV1)*T    !ND,Controller 1 Output

    If(iRes==9)E2(iCount)=SPxb-PV2
    B=Kc2*(SPxb-PV2)+Ki2*(SPxb-PV2)*T    !ND,Controller 2 Output

    F(1)=0
    F(2)=0
    F(3)=0
    F(4)=0
    F(5)=0
    F(6)=0

Case(1)          !ND,Analyzer Dead Time Ends
    PV1=QuadraticSpline(Time(1:iCount),Y1(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y2(1:iCount),T-5)    !ND,Process Var 1 Eqn
    PV2=QuadraticSpline(Time(1:iCount),Y4(1:iCount),T-
5)+QuadraticSpline(Time(1:iCount),Y5(1:iCount),T-5)    !ND,Process Var 2 Eqn

    If(iRes==9)E1(iCount)=SPxd-PV1          !ND,Trapezoid Rule for Integral Action of
Controller 1
    IE1=0.0
    Do i=1,iCount-1
    IE1=IE1+(E1(i+1)+E1(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE1=IE1+(E1(iCount)+SPxd-PV1)*(T-Time(iCount))/2.0

    L=Kc1*(SPxd-PV1)+Ki1*IE1          !ND,Controller 1 Output

    If(iRes==9)E2(iCount)=SPxb-PV2          !ND,Trapezoid Rule for Integral Action of
Controller 1
    IE2=0.0
    Do i=1,iCount-1
    IE2=IE2+(E2(i+1)+E2(i))*(Time(i+1)-Time(i))/2.0
    EndDo
    IE2=IE2+(E2(iCount)+SPxb-PV2)*(T-Time(iCount))/2.0

    B=Kc2*(SPxb-PV2)+Ki2*IE2          !ND,Controller 2 Output

    F(1)=-U(1)+KP11*L
    F(2)=-U(2)+KP12*B
    F(3)=-U(3)
    F(4)=-U(4)+KP21*L
    F(5)=-U(5)+KP22*B
    F(6)=-U(6)

End Select

@Coefficient Matrix
E(1)=Tau11          !ND,Tau's for differential eqns
E(2)=Tau12
E(3)=TauD1
E(4)=Tau21
E(5)=Tau22
E(6)=TauD2

@Solver Options
Neq=6
Npts=100

```

Tend=200.0
Tstop=5

Param(28)=1	!ND,Continue Integration Past Dead Time
Param(177)=1	!ND,Enable Integration History Access
VariableNames=Time;Y1;Y2;Y3;Y4;Y5;Y6	!ND,Variable Names
Append=PV1;PV2	!ND,Display Process Variable